BAEKSEOK UNIVERSITY

# C Syntax

# 목차

# 01 Syntax

# Syntax

```
1 #include <stdio.h>
2
3 int main() {
4   printf("Hello World!");
5   return 0;
6 }
```

Line 1: #include <stdio.h> tells C to include a header file. This header lets us use input/output functions such as printf() (used in line 4).

Line 2: A blank line. C ignores extra spaces and blank lines, but we use them to make the code easier to read.

Line 3: main() is a special function. Your program starts running here. Any code inside the curly brackets {} will be executed.

Line 4: printf() is a function used to output (print) text to the screen. In our example, it prints Hello World!.

Line 5: return 0 ends the main() function and sends a value back to the operating system.

Returning 0 usually means "everything worked". You will learn more about return values later.

Line 6: Do not forget the closing curly bracket } to end the main() function.

# C Statements

- A computer program is a list of instructions that a computer follows.
- In C, these instructions are called statements.
- Every statement in C must end with a semicolon ;.
  - If you forget the semicolon, the compiler will show an error and the program will not run:

```
printf("Hello World!")
```

```
error: expected ';' before 'return'
```

- Many Statements
  - Most C programs contain more than one statement.
  - Statements are executed one by one, in the same order as they are written:

# Challenge: Create a Simple C Program

- Instructions
  - Use the editor to complete the following steps:
  - Write a main() function
  - Use printf to print Hello, C!
  - End the program with return 0;

Code Solution

```
#include <stdio.h>

// 1) Write the main() function here
// 2) Use printf() to print "Hello, C!"
// 3) End the program with return 0;
// Note: Don't forget }
```

# C Output

# printf()

- Output (Print Text) : Double Quotes
  - To output values or print text in C, you can use the printf() function:
  - When you want to print text, the text must be wrapped in double quotes "".

- Example

```c
#include <stdio.h>

int main() {
  printf("Hello World!");
  return 0;
}
```

# New Lines

- You can use as many printf() functions as you want.
- However, note that it does not insert a new line at the end of the output:

```c
#include <stdio.h>

int main() {
  printf("Hello World!");
  printf("I am learning C.");
  printf("And it is awesome!");
  return 0;
}
```

# New Lines

- To start a new line when printing text, you can use the \n character:

```c
#include <stdio.h>

int main() {
  printf("Hello World!\n");
  printf("I am learning C.");
  return 0;
}
```

# New Lines

- You can also print multiple lines using a single printf() function. However, this could make the code harder to read:

```c
#include <stdio.h>

int main() {
  printf("Hello World!\nI am learning C.\nAnd it is awesome!");
  return 0;
}
```

```c
#include <stdio.h>

int main() {
  printf("Hello World!\n\n");
  printf("I am learning C.");
  return 0;
}
```

# New Lines

- \n is called an *escape sequence*.
- Here are some other common escape sequences:
  - \t : Inserts a horizontal tab
  - \\ : Inserts a backslash character (\)
  - \ " : Inserts a double quote character

# Output

- Challenge: Print Text Output
- Instructions
  - Inside main(), complete the following steps:

1. Use one printf to print Hello World! followed by a new line
2. Use another printf to print I am learning C.

  - Note: Remember to end a statement with semicolon.

Code Solution

```c
#include <stdio.h>

int main() {
  // 1) Print "Hello World!" with a new line
  // 2) Print "I am learning C."

  return 0;
}
```

# Comments

- Comments in C
  - Comments are notes for humans. They help explain code and make it easier to read.
  - They can also be used to temporarily "disable" code while testing.
  - Comments can be single-line or multi-line.
- Single-line Comments
  - Single-line comments start with two forward slashes (//).
  - Any text between // and the end of the line is ignored by the compiler.
  - This example uses a single-line comment before a line of code:
- Multi-line Comments
  - Multi-line comments start with /* and end with */.
  - Any text between /* and */ will be ignored by the compiler:

# Comments

- Example

```
1 printf("Hello World!"); // This is a comment
2
3 printf("This will run");
4 // printf("This will NOT run");
5
6 /* The code below will print the words Hello World!
7 to the screen, and it is amazing */
8 printf("Hello World!");
```

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# Thank You !

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